VELOOP REWARDS

Data Analyst Internship - Task 01

Product Analytics, Data Readiness & Data Management Blueprint

Final Consolidated Report (Deliverable 21)

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DATA AUTHENTICITY NOTICE

All datasets used are SIMULATED / FICTIONAL. No figure in this report is an actual VELOOP user, earning, withdrawal, revenue, retention, conversion or performance statistic. Events, fields and KPIs described as Proposed are not claimed to exist in production. Every simulated result is worded "In the simulated dataset...".

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01 Executive Summary

What was investigated

This project builds the analytical foundation for VELOOP Rewards, a rewards and engagement product that is still under development and does not yet have enough production data for conventional performance analysis. The work covers what VELOOP should collect, how it should be structured, how to know it is trustworthy, what management should measure and what engineering should implement. Two simulated datasets were examined: the Task 01 Users workbook (1,000 clean and 1,008 audit rows, 01 Apr - 15 Sep 2026) and a supplementary behavioral workbook labelled "Task 02" (sessions, events, tasks).

What was discovered

- The supplied Task 01 workbook contains the Users entity only. The six other datasets listed in the brief (Task Activity, Rewards, Wallet, Withdrawals, Referrals, Feature Events) were not supplied, so wallet, withdrawal, reward and referral KPIs could be designed but not tested (see section 12).
- The two files cannot be joined. All 1,000 user_id values overlap, but in the simulated data none of the emails, names, registration times or referral codes match. This identifier collision is the single most serious data finding (DQ-014).
- The Users audit sheet contains 8 exact duplicate rows, 9 rows with single missing values and 1 malformed registration date ("2026-14-02 12:00"). Formats, statuses and whitespace checks found no further problems.
- The "clean" sheet still shows state conflicts: 72 users with sessions but no login, 23 users with onboarding Completed and 0 sessions, 34 users with consent Pending but marketing opt-in Yes, 9 Suspended users flagged as active in the last 7 days, 4 negative derived "days since registration" values and 4 logins after the stated period end.
- In the supplementary simulated data, referral activity exists only as events (referral_created, shared, page viewed) with no referrer-to-referred link, and wallet activity is only wallet_viewed. Referral conversion and wallet reconciliation therefore cannot be measured from either file.
- **Output of the design work:** 42 proposed events, 106 data-dictionary fields over 10 entities, 54 KPIs, 44 dashboard visuals, 26 logged data-quality issues, 22 quality controls, 30 registered assumptions, 14 analytics gaps and a 22-item backlog.

What is currently unknown

No production implementation could be verified. In particular: whether events, event_id, session_id and wallet transaction_id exist (A001, A004, A005, A009); how reward units relate (A008); which withdrawal methods exist (A010); what activation and conversion mean for referrals (A013); and whether status history is stored (A011). All of these are recorded as To Be Verified rather than assumed.

What VELOOP should implement first

- Stable user_id, unique event_id with idempotent writes, and a UTC timestamp standard.
- An append-only wallet ledger with a stable transaction_id and linked reversals.
- Task, reward and withdrawal lifecycle events (task_completed only after validation; reward_credited only when the ledger entry exists).
- A referral table recording referrer, referred user and stage timestamps.
- An account status history table.
- A raw layer with validation and a quarantine table before any dashboard is built.

Major risks

- Duplicate or replayed events inflating every KPI (10 duplicate events exist in the supplementary audit sheet).
- Ledger and referral data that cannot be reconciled, which would undermine user trust in payouts.
- Third-party outcomes (ads, surveys, offers) being attributed to VELOOP rather than to the external system.
- Collecting more personal data than needed (the supplied Users sheet carries name, email, age and gender that no KPI requires).

- Reading simulated results as product performance.

Major priorities

Critical: user_id, event_id, transaction_id, referral linkage, withdrawal lifecycle timestamps, task and reward events. High: status history, session_id, approved enum lists, validation layer, KPI definitions. These are detailed in sections 19 and 21. The overall readiness picture is deliberately conservative: design completeness is assessed at an average of 3.3 out of 5, while verified production readiness is 0 out of 5 because no production evidence exists yet.

02 Scope, Sources and Data Authenticity

The work follows the integrity rules of the task brief (sections 2, 30 and 34): no fabricated production data, no simulated result presented as a VELOOP result, no invented security rules, no credentials, no unnecessary personal data, and no assumption converted into a fact. Every important product fact is recorded in the Source & Verification Register below (full register in workbook sheet 1).

ID	Information	Source Type	Status
S001	VELOOP is a developing rewards & engagement ecosystem without enough production data for conventional performance analysis	Official (task brief)	Confirmed
S002	Relevant product areas: accounts, earning activities, VEs, XP, Gems, Tokens, Spins, daily rewards, streaks, tasks, missions, badges, levels, referrals...	Official (task brief)	Confirmed
S003	Not every feature may be implemented or production-ready	Official (task brief)	Confirmed
S004	Task 01 Users dataset: 1,000 clean rows, 1,008 audit rows, period 01 Apr 2026 - 15 Sep 2026, 27 columns (28 in audit incl. data_environment)	Simulated	Confirmed
S005	Supplied Task 01 workbook contains the Users entity only; Task Activity, Rewards, Wallet, Withdrawals, Referrals and Feature Events sheets listed in b...	Simulated	Confirmed
S006	Supplementary behavioral dataset (Users 1,000; Sessions 7,913; Feature Events 35,704; Daily Activity 6,667; Task Activity 9,846; Milestones 4,799; Eve...	Simulated	Confirmed
S007	15 feature names appear in the supplementary dataset: Watch Ads, Missions, Streak Rewards, CAPTCHA, Tap & Earn, Mining Rewards, Referral, Daily Reward...	Simulated	To Be Verified
S008	Supported withdrawal methods, minimum amounts, review rules	n/a	To Be Verified
S009	Definitions and conversion between VEs, XP, Gems, Tokens	Official (task brief)	To Be Verified
S010	External providers behind ads, surveys, offers, games, tracking	n/a	To Be Verified
S011	Session identifier exists in production events	Official (task brief)	To Be Verified
S012	Definitions of derived flags is_active_7d, is_active_30d, days_since_registration and the reference timestamp used	Simulated	To Be Verified
S013	GPT-industry competitor patterns (rewards mechanics, withdrawal trust, tracking trust)	Internal (secondary research)	Proposed
S014	Data authenticity and integrity rules (no fabricated stats, no credentials, no unnecessary personal data)	Official (task brief)	Confirmed

Wording rule used throughout: results from simulated data are written "In the simulated dataset..."; proposed items are marked Proposed; unverified items are marked To Be Verified.

Supplementary dataset - how it was used

Because the Task 01 workbook holds only Users, the supplementary behavioral workbook was used as a clearly labelled test bed for relationship and KPI tests that would otherwise be impossible. It was never joined to the Task 01 Users sheet, and none of its results are mixed into Task 01 statistics. When the missing Task 01 sheets arrive, the tests in section 12 should be re-run on them (assumption A003).

03 Product Data Audit (A)

A1 Product area mapping and the A4 coverage matrix (31 rows) are in the workbook (sheets 3 and 4). The summary below groups them by area and priority.

Product Area	Rows	Features covered	Priorities
Account	7	Registration, Email verification, Login, Logout, Onboarding, Account status, Consent & marketing preference	Critical, High, Medium, Low
Tasks	5	Task discovery, Task open, Task start, Task completion, Task failure / abandonment	Critical, High, Medium
Ads	1	Watch Ads	High
Earning	1	Tap & Earn / Mining / CAPTCHA / Follow & Earn	Medium
Rewards	3	Reward credit, Reward pending, Reward reversal / correction	Critical, High
Engagement	6	Daily reward, Streaks, Missions, Spin the Wheel, Badges, Levels / XP, Leaderboard	High, Medium, Low
Referrals	2	Referral link/code, Referral registration & activation	Critical, High
Wallet	2	Wallet transactions, Exchange	Critical, Medium
Withdrawals	2	Withdrawal request, Withdrawal processing/completion/rejection/reversal	Critical
Support	2	Help center, Support request	Medium

Distinguishing user events, system events and business outcomes (A2, A3)

Class	Meaning	Examples (all Proposed)
User Event	Something the user does	task_started, daily_reward_claimed, withdrawal_requested, spin_started
System Event	Something the system does without a user action	reward_pending, reward_reversed, account_status_changed, wallet_transaction_created, task_failed
Business Outcome	A result with business meaning, reached after validation	task_completed, reward_credited, referral_converted, wallet_transaction_completed, withdrawal_completed

Example - task system: Task viewed > opened > started > completed > completion validated > reward credited. Measurable events: viewed, started, completed (after validation) and reward_credited. "Opened" is kept but Medium priority; "completion validated" is folded into task_completed, because emitting it earlier would count unvalidated completions.

Analytics gaps (10+ required; 14 identified)

ID	Gap	Priority	Reason
AG01	No verified stable event_id and no idempotency design	Critical	Without a unique event_id, duplicates cannot be removed; the supplementary audit file contains 10 duplicated event rows.
AG02	Wallet transaction_id and ledger design unverified	Critical	Without a stable transaction ID reliable deduplication and balance reconciliation cannot be guaranteed.
AG03	No referrer-to-referred link in supplied Users data (only "Friend/Family" label and one code per user)	Critical	Referral conversion cannot be computed without referrer and referred user IDs.
AG04	No account status-change history	High	Current status alone cannot show whether activity occurred before or after suspension (25 simulated suspended users show 30-day activity).
AG05	Reward credited vs activity occurred not separable in Task 01 data	Critical	No Rewards sheet was supplied; reward value cannot be reconciled with completions.
AG06	Withdrawal lifecycle timestamps not available	Critical	Processing time, pending queue and completion rate need requested/processing/final timestamps.
AG07	Session ID not verified in production	High	Funnel steps cannot be tied to one visit without a session

ID	Gap	Priority	Reason
			identifier.
AG08	Failure classification does not exist (only proposed)	High	Failure reasons need a controlled list agreed with Engineering.
AG09	Derived-field definitions undocumented (active flags, days since registration)	Medium	Flags could not be reproduced for 8-9 users; KPI definitions would drift.
AG10	Timezone and timestamp standard unknown	High	Mixed zones shift day boundaries; malformed dates exist in the audit sheet.
AG11	No event for onboarding milestones beyond three status values	Medium	Drop-off location cannot be found.
AG12	Third-party outcome signals not defined	High	External delays and reversals would be wrongly attributed to VELOOP.
AG13	Support/help-center tracking unknown	Medium	No source system identified.
AG14	Consent history not stored (current flag only)	Medium	Audit of consent timing impossible.

04 User & Account Analytics (B)

Stable identifiers needed (B1):

Identifier	Why needed
user_id	Immutable, non-personal key joining every entity; must never be email or name
event_id	De-duplicates replayed events
session_id	Groups events into a visit; existence unverified (A004)
transaction_id	Reconciles and de-duplicates wallet entries; links reversals (A009)
task_activity_id / reward_id / withdrawal_id / referral_id	One id per attempt, reward, request and referral so each lifecycle can be traced

The user analytics specification (B2) lists every Users field with type, required flag, purpose, source, validation and status (workbook sheet 6). Two fields, full_name and email, are deliberately excluded from the analytics layer: no KPI needs them, and the brief asks not to request unnecessary personal information. Where de-duplication is needed, a one-way hash of the email can be used.

Lifecycle measurement: registration (user_registered), email verification (email_verified), first login and returns (user_logged_in), onboarding (started/completed; step milestones await A026) and account status changes (account_status_changed with old status, new status, reason category and actor type).

Simulated observations (Task 01 Users, 1,000 rows)

In the simulated dataset: 855 of 1,000 users (85.5%) are Verified; 722 (72.2%) completed onboarding; 77 (7.7%) are Suspended; 923 (92.3%) accepted consent; Android is the primary device for 659 users (65.9%) and platform is Web for all. These figures illustrate that the KPIs can be computed; they are not VELOOP results.

05 Earning & Reward Analytics (C)

Lifecycle measured: Discovery (task_viewed) > Start (task_started) > Completion (task_completed, only after validation) > Validation (validation_status) > Reward (reward_pending, reward_credited). Each stage has an event in section 8.

C1 Task-level fields

task_activity_id, task_id, task_type, feature, user_id, started_at, completed_at, status, reward_amount, reward_unit, validation_status, provider_ref (source information). See dictionary entities Tasks, Task Activity and Rewards.

C2 Failed activity - Proposed classification

failure_category (Proposed)	Meaning	Stage	Observed in simulated data as
tracking_failed	External or internal tracking did not register the action	tracking	-
validation_failed	Completion was rejected by validation	validation	Validation failed
task_unavailable	Task could not be performed	availability	Task unavailable
timeout	Attempt expired	system	Timeout
incomplete_action	User did not finish required steps	user	Incomplete action
user_exited	User left	user	User exited
system_error	Application error	system	-

No internal failure codes are invented: the categories are marked Proposed and must be agreed with Engineering (AG08).

C3 Reward events

Activity occurred is not the same as reward credited. task_completed records the validated activity; reward_credited records the ledger credit and carries wallet_transaction_id. reward_pending, reward_reversed and reward_corrected record the other states. Reconciliation rule QC-16 compares completed attempts with credited rewards.

In the supplementary simulated data, all 5,627 Completed attempts have reward_status Credited and no non-completed attempt is Credited; this proves nothing about wallet credits because no Rewards or Wallet table exists in either file.

C4 Third-party dependency

For ads, surveys, offers and games an external system confirms completion, may delay it, and may later reverse it. The design therefore (i) flags tasks with is_third_party, (ii) stores provider_ref and provider outcome fields as optional properties, (iii) allows reward_pending and reward_reversed with a reason category, and (iv) reports third-party reversal rates separately so they are not blamed on VELOOP. Which providers exist is To Be Verified (A012).

Supplementary simulated observation

In the supplementary simulated data, 9,846 task attempts: 5,627 Completed, 648 Pending, 3,571 Failed or Abandoned. Completion is 57.2% of all attempts (61.2% excluding Pending); the spread across eight task categories is narrow (56.1% to 58.2%).

06 Wallet & Withdrawal Analytics (D)

D1 Wallet transaction model

Field	Notes
transaction_id	Unique, immutable; Critical
user_id, transaction_ts	Owner and UTC posting time
transaction_type	reward_credit, redemption, reversal, adjustment, exchange - ALL Proposed (A009, A024)
amount, unit	Signed amount; never aggregated across units
status	created, completed, reversed, failed
source, reference_id	Origin system and triggering record (e.g. reward_id)
related_transaction_id	Links a reversal/correction to the original entry

D2 Withdrawal lifecycle

Requested > Under review / Processing > Completed, with alternatives Rejected and Reversed (Reversed only after Completed). Fields: withdrawal_id, user_id, requested_amount, unit, method, requested_at, processing_at, final_at, status, reason_category. Valid transitions are enforced by QC-11; timestamps must be ordered (QC-12). Methods are not assumed (A010).

D3 Withdrawal KPIs

KPI	Definition	Formula	Frequency	Stakeholder	Limitation
Withdrawal requests	Count of requests created	COUNT(withdrawal_id)	Daily	Operations, Finance	Table not supplied
Completed withdrawals	Requests with status completed	COUNT WHERE status = completed	Daily	Operations	Table not supplied
Pending withdrawal queue	Requests requested/processing and not final	COUNT WHERE status IN (requested, processing)	Daily	Operations	Table not supplied
Rejected withdrawals	Requests ending rejected, by reason	COUNT WHERE status = rejected GROUP BY reason_category	Weekly	Operations	Table not supplied
Processing time	Time from request to final state	AVG/MEDIAN(final_at - requested_at) for terminal requests	Daily	Operations	Open requests excluded; table not supplied
Withdrawal volume	Sum of requested/completed amounts by unit	SUM(requested_amount) GROUP BY unit, status	Daily	Finance	Table not supplied
Method distribution	Share of requests by method	requests per method / total	Weekly	Product, Finance	Methods unverified (A010)
Reversed withdrawals	Completed withdrawals later reversed	COUNT WHERE status = reversed / completed	Weekly	Finance	Table not supplied
Withdrawal completion rate	Completed share of terminal requests	completed / (completed+rejected+reversed)	Weekly	Operations	Table not supplied

Each KPI also lists data required, source and purpose in the workbook (sheet 11). None can be calculated from the supplied files because no withdrawal data exists (section 12).

07 Referral & Engagement Analytics (E)

Funnel: Referral Created > User Registered > User Activated > Referral Converted. Registration is never treated as conversion. Activation and conversion definitions are open business decisions (A013); the events carry an activation_rule / conversion_rule property so the definition used is traceable.

Stage	Event	Timestamp field	KPI
Created	referral_created	created_at	K-F01
Registered	referral_registered	registered_at	K-F02
Activated	referral_activated	activated_at	K-F03
Converted	referral_converted	converted_at	K-F04, K-F05

Finding: in the simulated Users data a referral user has only the label "Friend/Family", and every user has a referral code; there is no referrer user_id. Referral attribution is therefore impossible (DQ-021, critical gap AG03). The supplementary events contain 871 referral_created events, 834 shares and 816 page views, with no linking field.

Engagement measurement: feature adoption (K-E14), repeat usage (K-E15), task activity (K-E01..E05), daily activity and active users (K-U03, K-U04), returning users (K-U06). DAU is defined as unique users generating a qualifying activity event within a calendar day; the qualifying-event list is open (A017). In the supplementary simulated events, the average number of distinct users with any event per day is 40.7 (min 1, max 164).

08 Event Tracking Plan and Naming Standard (F)

Naming standard: object_action, snake_case, past-tense verb (e.g. user_registered, task_completed). Reasons: it matches the brief minimum events; it is short and sorts related events together; the feature is carried as a property so the name set stays small; an event is emitted only after the state is true. Properties use snake_case, enums lower case, amounts decimal with a separate unit, timestamps UTC ISO-8601; no personal data in properties. All 42 events follow it.

Event	Class	Domain	Priority	Status
user_registered	User Event	account	Critical	Proposed
email_verified	User Event	account	High	Proposed
user_logged_in	User Event	account	Critical	Proposed
user_logged_out	User Event	account	Low	Proposed
onboarding_started	User Event	account	High	Proposed
onboarding_completed	User Event	account	High	Proposed
account_status_changed	System Event	account	High	Proposed
consent_updated	System Event	account	Medium	Proposed
task_viewed	User Event	tasks	High	Proposed
task_opened	User Event	tasks	Medium	Proposed
task_started	User Event	tasks	Critical	Proposed
task_completed	Business Outcome	tasks	Critical	Proposed
task_failed	System Event	tasks	High	Proposed
task_abandoned	User Event	tasks	High	Proposed
reward_credited	Business Outcome	rewards	Critical	Proposed
reward_pending	System Event	rewards	High	Proposed
reward_reversed	System Event	rewards	High	Proposed
reward_corrected	System Event	rewards	Medium	Proposed
daily_reward_claimed	User Event	engagement	High	Proposed
streak_updated	System Event	engagement	Medium	Proposed
mission_started	User Event	engagement	Medium	Proposed
mission_completed	Business Outcome	engagement	Medium	Proposed
badge_earned	System Event	engagement	Low	Proposed
level_increased	System Event	engagement	Low	Proposed
spin_started	User Event	engagement	Medium	Proposed
spin_completed	Business Outcome	engagement	Medium	Proposed
feature_viewed	User Event	engagement	Medium	Proposed
referral_created	User Event	referrals	High	Proposed
referral_registered	System Event	referrals	High	Proposed
referral_activated	Business Outcome	referrals	High	Proposed
referral_converted	Business Outcome	referrals	Critical	Proposed
wallet_transaction_created	System Event	wallet	Critical	Proposed
wallet_transaction_completed	Business Outcome	wallet	Critical	Proposed
wallet_transaction_reversed	System Event	wallet	Critical	Proposed
exchange_completed	User Event	wallet	Medium	Proposed
withdrawal_requested	User Event	withdrawals	Critical	Proposed
withdrawal_processing	System Event	withdrawals	High	Proposed
withdrawal_completed	Business Outcome	withdrawals	Critical	Proposed
withdrawal_rejected	Business Outcome	withdrawals	High	Proposed
withdrawal_reversed	Business Outcome	withdrawals	High	Proposed
help_article_viewed	User Event	support	Medium	Proposed
support_ticket_created	User Event	support	Medium	Proposed

Full catalogue with trigger, definition, required/optional properties and types, validation, source, business purpose, KPI dependency and the name observed in simulated data: workbook sheet 7. None of the events is claimed to exist in production.

09 Data Dictionary (G)

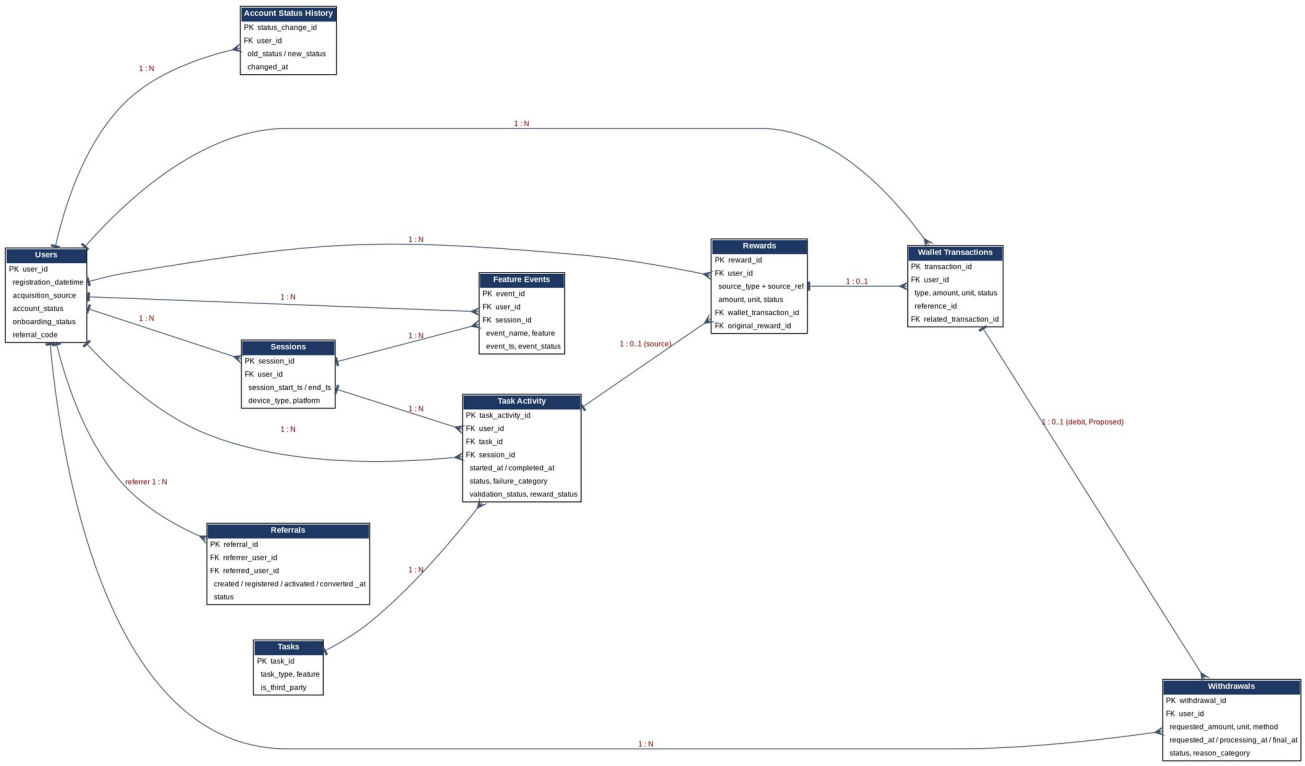
106 fields across 10 entities - above the 75 minimum - and each is meaningful (no filler). 44 fields exist in the simulated structures (marked "Confirmed (simulated structure)"); the rest are Proposed. Every field lists entity, name, type, definition, example, required flag, allowed values, validation, source, sensitivity, analytical use and status (workbook sheet 10).

Entity	Fields
Users	27
Account Status History	6
Sessions	8
Tasks	5
Task Activity	11

Entity	Fields
Rewards	11
Wallet Transactions	11
Withdrawals	10
Referrals	9
Feature Events	8

Sensitivity: Restricted = name, email, age, gender and consent attributes; Internal = everything else; no Public fields. No passwords, keys or payment credentials appear anywhere.

10 Logical Data Model (H)



Primary keys (H1): users.user_id; account_status_history.status_change_id; sessions.session_id; tasks.task_id; task_activity.task_activity_id; rewards.reward_id; wallet_transactions.transaction_id; withdrawals.withdrawal_id; referrals.referral_id; feature_events.event_id.

Foreign keys and relationships (H2, H3):

Relationship	Cardinality	Notes
Users - Account Status History	1 : N	Each status change is a row
Users - Sessions / Task Activity / Rewards / Wallet Transactions / Withdrawals / Feature Events	1 : N	user_id on every child
Users - Referrals	1 : N (referrer) and 1 : 0..1 (referred)	Two foreign keys to Users: referrer_user_id and referred_user_id; a user is referred at most once (not drawn to keep the diagram readable)
Tasks - Task Activity	1 : N	Task catalogue entry attempted many times
Sessions - Task Activity / Feature Events	1 : N	Optional where session_id is unavailable (A004)
Task Activity - Rewards	1 : 0..1	Rewards.source_type + source_ref points to the activity; other sources (daily reward, spin) use other tables
Rewards - Wallet Transactions	1 : 0..1	A pending reward has no ledger entry yet
Wallet Transactions - Wallet Transactions	1 : N (self)	related_transaction_id links reversals/corrections to the original
Wallet Transactions - Withdrawals	1 : 0..1 (Proposed)	Debit entry for a payout; depends on A009
Many-to-many	none stored directly	Users and Missions/Badges would be many-to-many through a link table if those entities are added; not included because the first release does not require them

Event relationships: every feature_event may reference task_activity, mission or spin by identifier in its properties; transaction relationships follow the ledger rules above. Missions, Badges, Levels and Support Tickets are deliberately kept out of the first model: their business value is lower and their definitions are unverified.

11 KPI Catalogue (I)

54 KPIs across User, Engagement, Reward, Wallet, Withdrawal, Referral and Support areas (minimum 30). Each has name, definition, formula, required fields, data source, frequency, stakeholder, business purpose, interpretation, limitation and status (workbook sheet 11). No fictional company targets are set: for example, DAU is defined as unique users generating a qualifying activity event within a calendar day, with no target value.

KPI ID	Area	Name	Formula	Testable?
K-U01	User	New registrations	COUNT(DISTINCT user_id) WHERE registration_ts in period	Yes: Task 01 Users sheet
K-U02	User	Registrations by acquisition source	registrations per source / total registrations	Yes: Task 01 Users sheet
K-U03	User	Daily Active Users (DAU)	COUNT(DISTINCT user_id) with >=1 qualifying event on date D	Yes: supplementary dataset only
K-U04	User	Monthly Active Users (MAU)	COUNT(DISTINCT user_id) WHERE event_ts in last 30 days	Yes: supplementary dataset only
K-U05	User	Email verification rate	verified users / registered users	Yes: Task 01 Users sheet
K-U06	User	Returning users	COUNT(DISTINCT user_id) active in period AND registration_ts < period start	Yes: supplementary dataset only
K-U07	User	Onboarding completion rate	users with onboarding_status = Completed / registered users	Yes: Task 01 Users sheet
K-U08	User	Suspended account share	Suspended / all accounts	Yes: Task 01 Users sheet
K-U09	User	Status transitions	COUNT(*) GROUP BY old_status, new_status	No: source table not supplied
K-U10	User	Registrations by device and platform	registrations per device / total	Yes: Task 01 Users sheet
K-U11	User	Consent acceptance rate	Accepted / all users	Yes: Task 01 Users sheet
K-U12	User	Inactive users	registered users - K-U04 users	Yes: Task 01 Users sheet
K-E01	Engagement	Task starts	COUNT(task_started)	Yes: supplementary dataset only
K-E02	Engagement	Task view-to-start rate	started attempts / viewed tasks	Yes: supplementary dataset only
K-E03	Engagement	Task completion rate	Completed attempts / (started attempts excluding Pending)	Yes: supplementary dataset only
K-E04	Engagement	Task completions by feature	COUNT(Completed) GROUP BY task_type	Yes: supplementary dataset only
K-E05	Engagement	Task failure and abandonment rate	(Failed+Abandoned)/ non-pending attempts	Yes: supplementary dataset only
K-E06	Engagement	Failures by category	COUNT(Failed) GROUP BY category	Yes: supplementary dataset only
K-E07	Engagement	Daily reward claim rate	daily claimers / DAU	Yes: supplementary dataset only
K-E08	Engagement	Daily claimers	COUNT(DISTINCT user_id)	Yes: supplementary dataset only
K-E09	Engagement	Average session duration	AVG(session_end - session_start)	Yes: supplementary dataset only
K-E10	Engagement	Streak continuation rate	extended / (extended+broken)	Yes: supplementary dataset only
K-E11	Engagement	Mission completion rate	mission_completed / mission_started	Yes: supplementary dataset only
K-E12	Engagement	Badge and level progression	events / DAU	Yes: supplementary dataset only
K-E13	Engagement	Spin usage	COUNT(DISTINCT user_id) spin_started	Yes: supplementary dataset only
K-E14	Engagement	Feature usage	COUNT(DISTINCT user_id) GROUP BY feature	Yes: supplementary dataset only
K-E15	Engagement	Repeat usage	users with >=2 active days in feature / feature users	Yes: supplementary dataset only
K-R01	Reward	Reward credits	COUNT/SUM(amount) WHERE status = credited, by unit	No: source table not supplied
K-R02	Reward	Reward activity by feature	SUM(amount) GROUP BY source_type, unit	No: source table not supplied
K-R03	Reward	Pending reward queue	COUNT pending; AVG(now - created_at)	No: source table not supplied
K-R04	Reward	Reward reversal rate	COUNT reversed / COUNT credited	No: source table not supplied
K-R05	Reward	Credit reconciliation	completed attempts with reward_status Credited / completed attempts	Yes: supplementary dataset only
K-W01	Wallet	Wallet transactions	COUNT(transaction_id) GROUP BY type	No: source table not supplied
K-W02	Wallet	Credit volume	SUM(amount) WHERE	No: source table not

KPI ID	Area	Name	Formula	Testable?
			type = reward_credit AND status = completed	supplied
K-W03	Wallet	Redemption activity	COUNT/SUM WHERE type = redemption	No: source table not supplied
K-W04	Wallet	Wallet reversal rate	COUNT reversed / COUNT completed	No: source table not supplied
K-W05	Wallet	Exchange usage	COUNT(DISTINCT user_id), SUM(from_amount)	No: source table not supplied
K-D01	Withdrawal	Withdrawal requests	COUNT(withdrawal_id)	No: source table not supplied
K-D02	Withdrawal	Completed withdrawals	COUNT WHERE status = completed	No: source table not supplied
K-D03	Withdrawal	Pending withdrawal queue	COUNT WHERE status IN (requested, processing)	No: source table not supplied
K-D04	Withdrawal	Rejected withdrawals	COUNT WHERE status = rejected GROUP BY reason_category	No: source table not supplied
K-D05	Withdrawal	Processing time	AVG/MEDIAN(final_at - requested_at) for terminal requests	No: source table not supplied
K-D06	Withdrawal	Withdrawal volume	SUM(requested_amount) GROUP BY unit, status	No: source table not supplied
K-D07	Withdrawal	Method distribution	requests per method / total	No: source table not supplied
K-D08	Withdrawal	Reversed withdrawals	COUNT WHERE status = reversed / completed	No: source table not supplied
K-D09	Withdrawal	Withdrawal completion rate	completed / (completed+rejected+reversed)	No: source table not supplied
K-F01	Referral	Referrals created	COUNT(referral_id)	No: source table not supplied
K-F02	Referral	Referral registrations	COUNT WHERE registered_at IS NOT NULL	No: source table not supplied
K-F03	Referral	Referral activations	COUNT WHERE activated_at IS NOT NULL	No: source table not supplied
K-F04	Referral	Referral conversions	COUNT WHERE converted_at IS NOT NULL	No: source table not supplied
K-F05	Referral	Referral stage conversion	stage n+1 count / stage n count	No: source table not supplied
K-S01	Support	Support requests	COUNT(ticket_id)	No: source table not supplied
K-S02	Support	Help-center engagement	help_article_viewed / DAU	No: source table not supplied
K-S03	Support	Issue categories	COUNT GROUP BY category	No: source table not supplied

12 Simulated Data Profiling (J)

All results below are from SIMULATED data. T01 = Task 01 workbook; SUP = supplementary workbook.

J1 Dataset profiles (full column-level profile in workbook sheets 12-13):

Dataset	Records	Cols	Date range	Unique IDs / dup IDs	Missing cells
T01 Users_1000_Clean	1,000	27	2026-04-01 to 2026-09-15	1,000 / 0	953 (referral_source_detail 881, last_login 72)
T01 Users_Messy_Audit	1,008	28	2026-04-01 to 2026-09-15	1,000 / 8 repeated rows	966 incl. injected defects
SUP Users	1,000	16	2026-04-01 to 2026-09-15	1,000 / 0	846 (referral detail)
SUP Sessions	7,913	12	2026-04-02 to 2026-09-15	7,913 / 0	466
SUP Feature Events	35,704	13	2026-04-02 to 2026-09-15	35,704 / 0	0
SUP Daily Activity	6,667	17	2026-04-02 to 2026-09-15	6,667 (user+date) / 0	0
SUP Task Activity	9,846	12	2026-04-02 to 2026-09-15	9,846 / 0	10,494 (structural nulls)
SUP Milestones	4,799	6	2026-04-02 to 2026-09-15	n/a	40
SUP Event audit sheet	35,714	13	2026-04-02 to 2026-09-15	35,704 / 10	0

Status distributions (T01 Users): account_status Active 568, Inactive 283, Suspended 77, Pending Verification 72; onboarding Completed 722, Started 146, Not Started 132; acquisition Organic Search 244, Direct 217, Social Media 160, Referral 119, WhatsApp 103, Other 102, Telegram 55. Numeric ranges: age 18-38; lifetime_sessions 0-81; active_days 0-32; xp_balance 5-1,156.

J2 Relationship tests (full list: workbook sheet 14):

Test	Result
T01 Users keys (user_id, email, referral_code) unique	Pass
T01 Users to Task Activity / Rewards / Wallet / Withdrawals / Referrals	Not testable - sheets not supplied
T01 Audit user_ids exist in Clean	Pass (8 repeated rows)
SUP Users to Sessions, Events, Tasks, Daily, Milestones	Pass - 0 orphans in all five
SUP: 16 users have no sessions and no tasks	Observation
T01 Users vs SUP Users on user_id	FAIL - same IDs, 0 matching emails/names/registration times: do not join (DQ-014)

J3 KPI test - can the KPI actually be calculated?

KPI	Calculable?	Missing data if not
K-U01, U02, U05, U07, U08, U10, U11 (user KPIs)	Yes - T01 Users	Timestamps for verification, consent, status changes
K-U03 DAU	Yes - SUP events only	Qualifying-event list; T01 has no events
K-E01, E03, E05 (task KPIs)	Yes - SUP only	Verified events and failure categories
K-R05 Credit reconciliation	Partly - SUP only	Rewards and Wallet tables
K-R01..R04, K-W01..W05	No	Rewards and Wallet Transactions tables
K-D01..D09	No	Withdrawals with lifecycle timestamps
K-F01..F05	No	Referrals with referrer and referred IDs
K-S01..S03	No	Support source

Selected simulated results: in the simulated dataset (T01) 855 of 1,000 are verified and 722 completed onboarding; in the supplementary simulated data 61.2% of non-pending task attempts completed. These show the KPIs are computable; they are not performance claims.

13 Data Quality Audit and Control Framework (K, L)

K - Data quality report (26 issues; excerpt, full table in workbook sheet 16)

ID	Dataset / field	Issue	Records	Sev.	Status
DQ-001	Users_Messy_Audit / user_id (also email, referral_code)	Duplicate records	8 rows (8 user_ids appear twice; exact duplicates)	High	Proposed
DQ-002	Users_Messy_Audit / city	Missing value	1 (VLUSR000001)	Medium	Proposed
DQ-003	Users_Messy_Audit / acquisition_source	Missing value	1 (VLUSR000002)	High	Proposed
DQ-004	Users_Messy_Audit / primary_device	Missing value	1 (VLUSR000003)	Medium	Proposed
DQ-005	Users_Messy_Audit / onboarding_status	Missing value	1 (VLUSR000004)	Medium	Proposed
DQ-006	Users_Messy_Audit / email_verification_status	Missing value	1 (VLUSR000006)	Medium	Proposed
DQ-007	Users_Messy_Audit / age	Missing value	1 (VLUSR000007)	Low	Proposed
DQ-008	Users_Messy_Audit / gender	Missing value	1 (VLUSR000008)	Low	Proposed
DQ-009	Users_Messy_Audit / registration_datetime	Missing timestamp	1 (VLUSR000023)	High	Proposed
DQ-010	Users_Messy_Audit / registration_datetime	Malformed date	1 (VLUSR000024: "2026-14-02 12:00")	High	Proposed
DQ-011	Users_1000_Clean / days_since_registration	Invalid derived value (negative)	4 (VLUSR000128, 000684, 000686, 000703)	Medium	Proposed
DQ-012	Users_1000_Clean / is_active_7d / is_active_30d	Derived flag not reproducible	8 (7d) and 9 (30d) users differ from login-based rule at reference 15 Sep 2026	Medium	Proposed
DQ-013	Users_1000_Clean / last_login_datetime	Date outside stated period	4 logins after 15 Sep 2026 (up to 17 Sep): VLUSR000128, 000665, 000686, 000703	Medium	Proposed
DQ-014	Users (Data_01) vs Users (Task 02 file) / user_id	Identifier collision across files	1,000 shared IDs; 0 matching emails/names/registration times	Critical	Open
DQ-015	Users_1000_Clean / last_login_datetime vs lifetime_sessions	Conflicting state	72 users with sessions > 0 and no login (all Pending Verification)	Medium	Open
DQ-016	Users_1000_Clean / onboarding_status vs lifetime_sessions	Impossible/unclear state	23 users Completed onboarding with 0 sessions	Medium	Open
DQ-017	Users_1000_Clean / account_status vs activity flags	Unexplained activity after suspension?	9 Suspended users flagged active 7d; 25 active 30d	High	Open
DQ-018	Users_1000_Clean / consent_status vs marketing_opt_in	Rule conflict	34 users consent Pending but marketing_opt_in = Yes	High	Open
DQ-019	Users_1000_Clean / email_verification_status vs account_created_via	Inconsistency to verify	40 Google Sign-In users with Pending verification	Low	Open
DQ-020	Users_1000_Clean / full_name	Non-unique attribute	596 repeated names	Low	Proposed
DQ-021	Users_1000_Clean / referral_source_detail	Design gap	Only value is "Friend/Family" for 119 referral users; no referrer user_id	High	Open
DQ-022	10_Event_Data_Quality_Audit (Task 02 file) / event_id	Duplicate events	10 duplicated event rows	High	Proposed
DQ-023	10_Event_Data_Quality_Audit / event_status	Trailing space and capitalization	4 rows ("success ")	Medium	Proposed
DQ-024	10_Event_Data_Quality_Audit / feature_name	Inconsistent naming/capitalization	5 rows (missions, leaderboard, wallet, daily reward, watch ads)	Medium	Proposed
DQ-025	02_Sessions vs 03_Feature_Events (Task 02 file) / events_in_session	Aggregate not matching detail	7,537 of 7,913 sessions differ from event row count	Medium	Open
DQ-026	05_Task_Activity (Task 02 file) / user_id vs account_status	Business-rule question	855 attempts by Pending Verification users; 601 by Suspended users	High	Open

Checks performed with no issue found:

- **Missing user_id:** 0 records (Users_Messy_Audit)
- **Invalid status values (Users):** 0 values outside controlled lists
- **Trailing spaces (Users):** 0 fields
- **Inconsistent capitalization (Users):** 0 fields
- Invalid numeric values / negatives (Users source fields): 0 (only the derived field in DQ-011)
- **Orphan records (Task 02 child tables):** 0 in sessions, events, tasks, daily, milestones
- **Impossible task transitions (Task 02):** 0 (completed without completion time; credited but not completed; completion before start)
- Invalid emails / user_id / referral_code format (Users): 0

Where the checklist in the brief found nothing (for example trailing spaces, invalid statuses and orphan records in Users) the result is reported as zero rather than inflated. Detection logic, likely cause, correction, prevention and owner for each issue are in the workbook.

L - Quality control framework

Controls are placed at the layer where each defect is cheapest to stop: Application > Data Collection > Raw Data > Transformation > Analytics Layer > Dashboard.

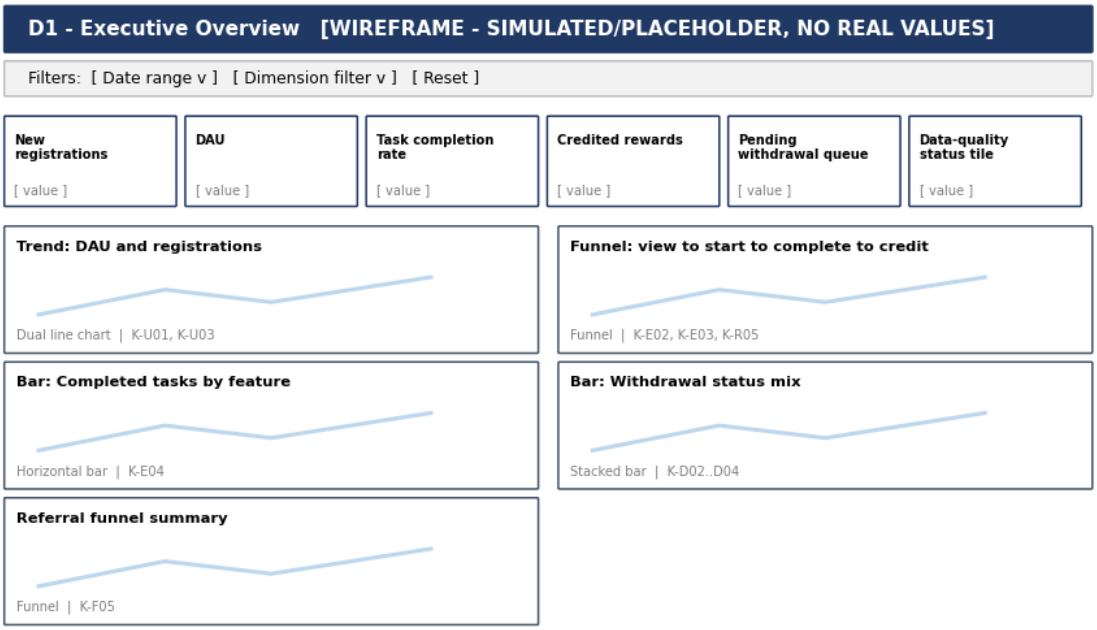
Rule	Quality rule	Layer	Action on failure
QC-01	Required fields present	Application	Reject with error; do not write
QC-02	ISO-8601 UTC timestamps	Data Collection	Quarantine event
QC-03	Approved status and enum values	Data Collection	Reject or map to Unknown and alert
QC-04	Unique event_id (idempotency key)	Data Collection	Drop duplicate; count it
QC-05	Unique transaction_id	Data Collection	Reject write
QC-06	Preserve raw data unchanged	Raw Data	Corrections in transformation only
QC-07	Schema drift detection	Raw Data	Alert; hold load
QC-08	Duplicate detection	Transformation	Keep first; log removed
QC-09	Trim and normalise text	Transformation	Apply mapping; log changes
QC-10	Referential integrity	Transformation	Move orphans to quarantine table
QC-11	Valid state transitions	Transformation	Flag record
QC-12	Timestamp order	Transformation	Flag record
QC-13	Derived-field recomputation	Transformation	Replace supplied derived values
QC-14	No cross-unit totals	Analytics Layer	Block metric
QC-15	Ledger reconciliation	Analytics Layer	Alert Finance
QC-16	Reward-activity reconciliation	Analytics Layer	Report gap
QC-17	Consent-marketing rule	Analytics Layer	Treat as No
QC-18	Dataset labelling	Analytics Layer	Block load
QC-19	Freshness	Dashboard	Show stale banner
QC-20	Row-count anomaly	Dashboard	Annotate dashboard; alert
QC-21	Quality tile	Dashboard	Warn viewers
QC-22	Suspended/pending account activity report	Analytics Layer	Report to Operations

14 Dashboard Blueprint (M)

Seven dashboards are specified (44 visuals). For each visual the workbook (sheet 18) records metric, calculation, source, visualization, filters, dimensions, date range, refresh frequency, business question and decision supported. Wireframes below use placeholders only; no values are shown because no production data exists.

D1 Executive Overview

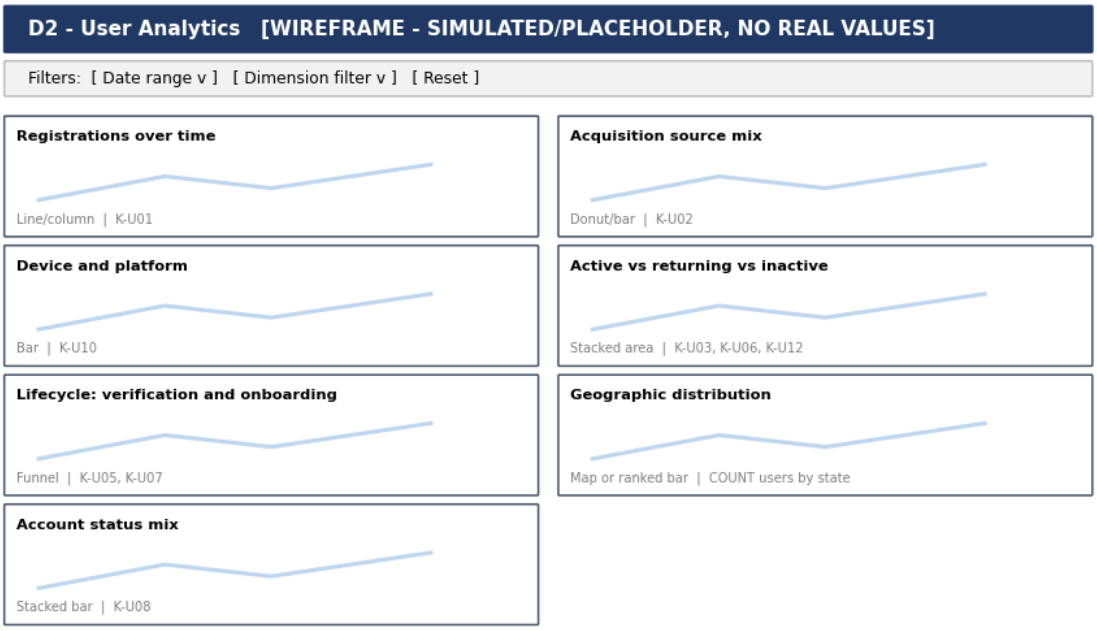
Audience: Executive, Product, Operations. Filters: All filters: date range, acquisition_source, platform. Date range: Last 30 days default; 7/30/90 and custom. Refresh: Daily.



Metric	KPI	Source	Visualization	Decision supported
KPI tile: New registrations	K-U01	Users / user_registered	KPI tile with change vs prior period	Adjust acquisition effort
KPI tile: DAU	K-U03	Event stream	KPI tile + sparkline	Judge product traction
KPI tile: Task completion rate	K-E03	Task Activity	KPI tile	Prioritise fixes
KPI tile: Credited rewards	K-R01	Rewards	KPI tile by unit	Reward budget
KPI tile: Pending withdrawal queue	K-D03	Withdrawals	KPI tile with alert colour	Staffing
Trend: DAU and registrations	K-U01, K-U03	Users, events	Dual line chart	Campaign timing
Funnel: view to start to complete to credit	K-E02, K-E03, K-R05	Task Activity, Rewards	Funnel	Roadmap
Bar: Completed tasks by feature	K-E04	Task Activity	Horizontal bar	Investment
Bar: Withdrawal status mix	K-D02..D04	Withdrawals	Stacked bar	Policy review
Referral funnel summary	K-F05	Referrals	Funnel	Referral reward design
Data-quality status tile	Quality checks pass rate	Quality log	KPI tile	Whether to publish

D2 User Analytics

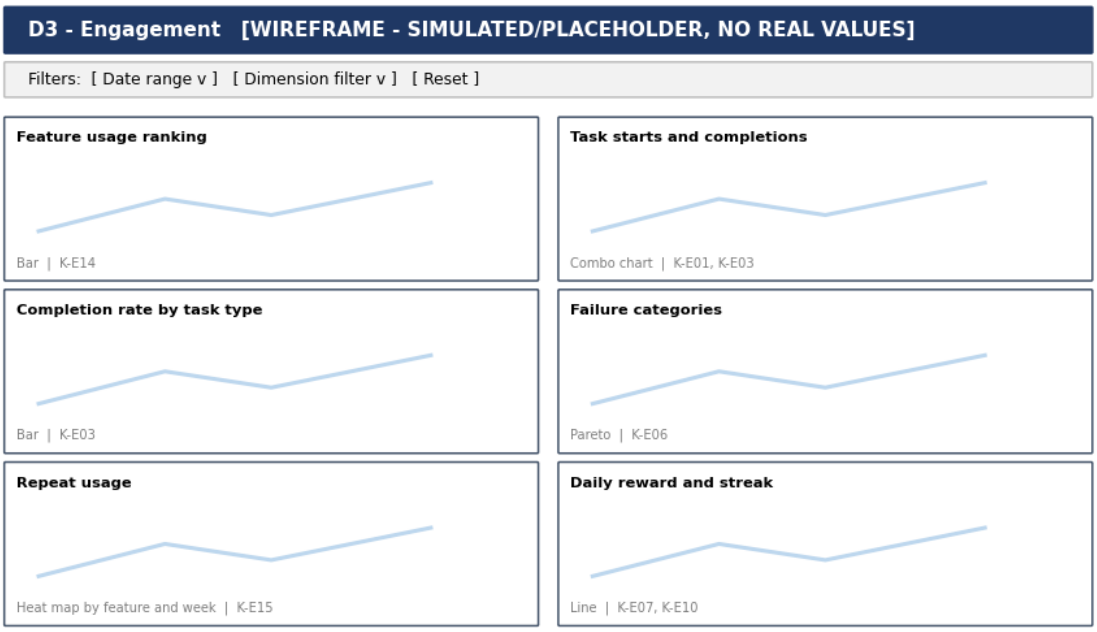
Audience: Product, Growth. Filters: Date range, source, device, state. Date range: Last 90 days; cohort by registration month. Refresh: Daily.



Metric	KPI	Source	Visualization	Decision supported
Registrations over time	K-U01	Users	Line/column	Marketing
Acquisition source mix	K-U02	Users	Donut/bar	Channel mix
Device and platform	K-U10	Users	Bar	Engineering priority
Active vs returning vs inactive	K-U03, K-U06, K-U12	Users, events	Stacked area	Re-engagement
Lifecycle: verification and onboarding	K-U05, K-U07	Users	Funnel	Onboarding changes
Geographic distribution	COUNT users by state	Users	Map or ranked bar	Regional focus
Account status mix	K-U08	Users	Stacked bar	Operations

D3 Engagement

Audience: Product. Filters: Date range, feature, task_type, device. Date range: Last 30 days. Refresh: Daily.

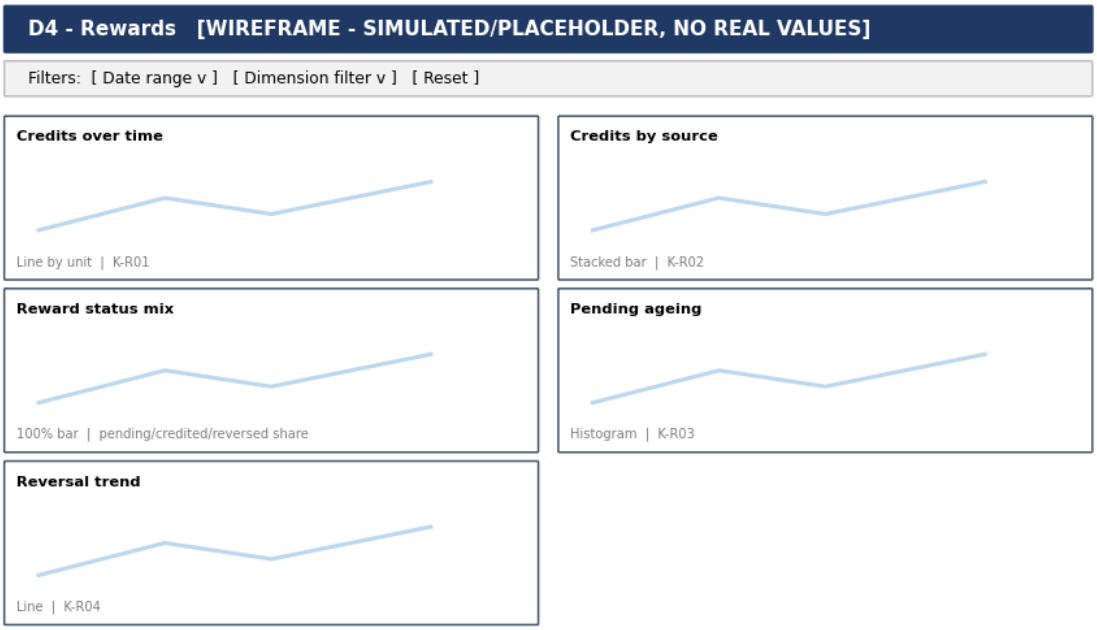


Metric	KPI	Source	Visualization	Decision supported
Feature usage ranking	K-E14	Events	Bar	Roadmap
Task starts and completions	K-E01, K-E03	Task Activity	Combo chart	Capacity
Completion rate by task type	K-E03	Task Activity	Bar	Task retirement

Metric	KPI	Source	Visualization	Decision supported
Failure categories	K-E06	Task Activity	Pareto	Bug priority
Repeat usage	K-E15	Events	Heat map by feature and week	Habit design
Daily reward and streak	K-E07, K-E10	Events	Line	Mechanic tuning

D4 Rewards

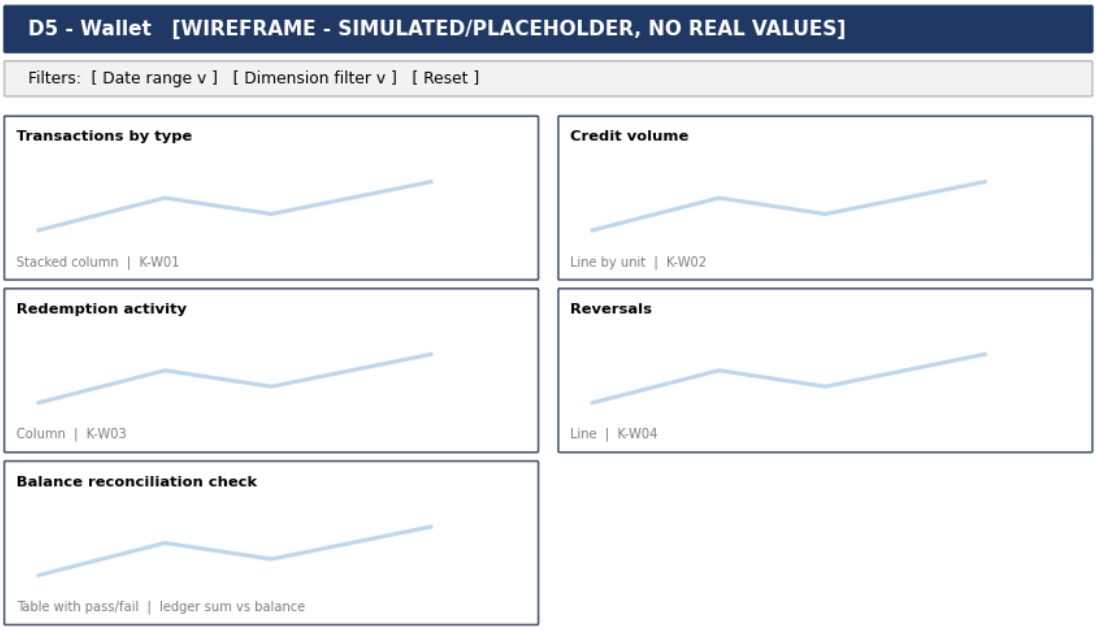
Audience: Product, Finance. Filters: Date range, source_type, unit, status. Date range: Last 30 days. Refresh: Daily.



Metric	KPI	Source	Visualization	Decision supported
Credits over time	K-R01	Rewards	Line by unit	Budget
Credits by source	K-R02	Rewards	Stacked bar	Rebalancing
Reward status mix	pending/credited/reversed share	Rewards	100% bar	Operations
Pending ageing	K-R03	Rewards	Histogram	Escalation
Reversal trend	K-R04	Rewards	Line	Provider review

D5 Wallet

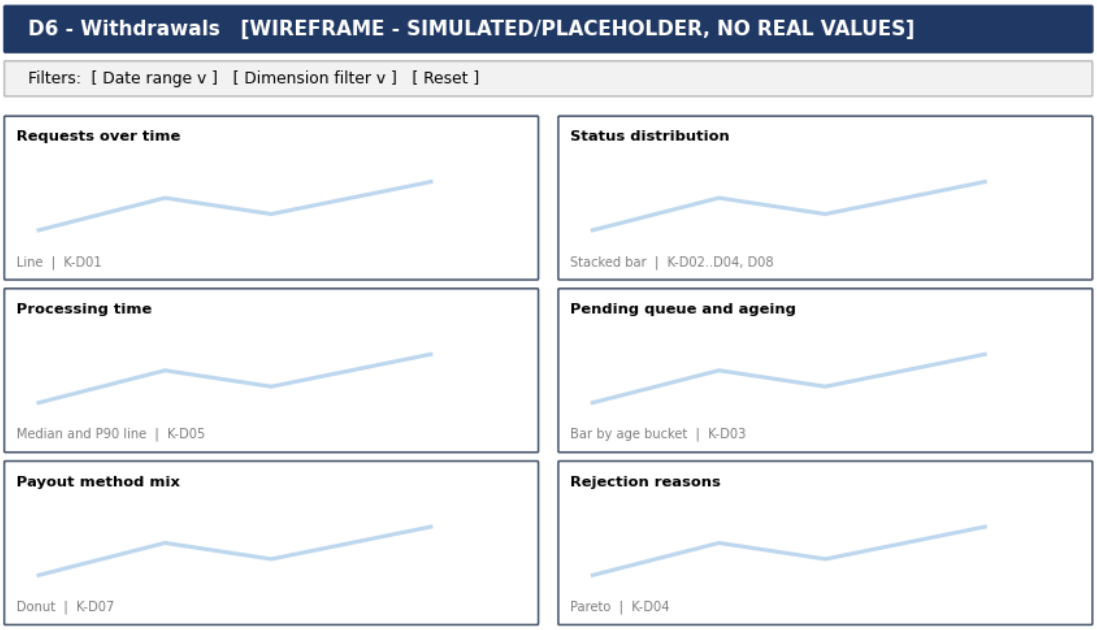
Audience: Finance. Filters: Date range, transaction_type, unit, status. Date range: Last 30 days. Refresh: Daily.



Metric	KPI	Source	Visualization	Decision supported
Transactions by type	K-W01	Wallet Transactions	Stacked column	Ledger health
Credit volume	K-W02	Wallet Transactions	Line by unit	Finance planning
Redemption activity	K-W03	Wallet Transactions	Column	Redemption design
Reversals	K-W04	Wallet Transactions	Line	Investigation
Balance reconciliation check	ledger sum vs balance	Wallet Transactions	Table with pass/fail	Release gate

D6 Withdrawals

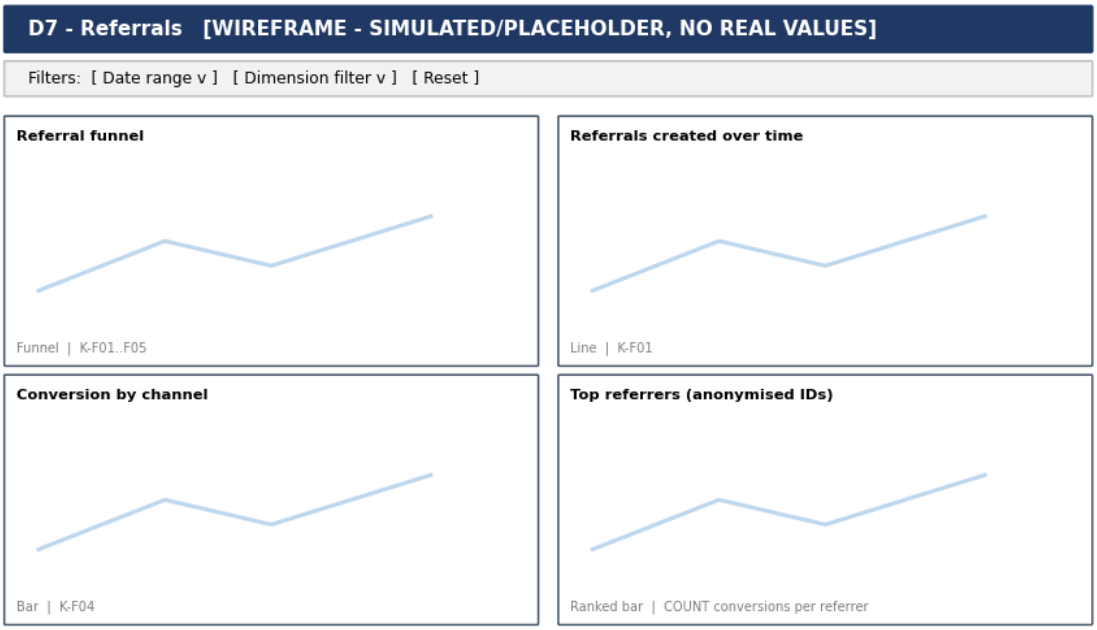
Audience: Operations, Finance. Filters: Date range, method, status. Date range: Last 30 days. Refresh: Hourly for queue; daily for rest.



Metric	KPI	Source	Visualization	Decision supported
Requests over time	K-D01	Withdrawals	Line	Capacity
Status distribution	K-D02..D04, D08	Withdrawals	Stacked bar	Policy
Processing time	K-D05	Withdrawals	Median and P90 line	SLA
Pending queue and ageing	K-D03	Withdrawals	Bar by age bucket	Staffing
Payout method mix	K-D07	Withdrawals	Donut	Method roadmap
Rejection reasons	K-D04	Withdrawals	Pareto	Message and rules

D7 Referrals

Audience: Growth. Filters: Date range, channel, stage. Date range: Last 90 days. Refresh: Daily.



Metric	KPI	Source	Visualization	Decision supported
Referral funnel	K-F01..F05	Referrals	Funnel	Referral design
Referrals created over time	K-F01	Referrals	Line	Campaigns
Conversion by channel	K-F04	Referrals	Bar	Spend
Top referrers (anonymised IDs)	COUNT conversions per referrer	Referrals	Ranked bar	Abuse review

15 Developer Analytics Handoff (N)

Fourteen critical events are specified with requirement ID, trigger, payload (field, type, required, validation), deduplication rule, timestamp rule, source, reason needed, KPI dependency, acceptance test and priority. The complete one-row-per-field table (108 rows) is in workbook sheet 9. Implementation is subject to final engineering design and idempotency decisions.

REQ-001 user_registered

Item	Specification
Trigger	Account record is successfully created
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); registration_method (enum: web_form, google_sign_in), acquisition_source (enum), device_type (enum), platform (enum)
Optional	referral_code_used (string), campaign_ref (string)
Validation	user_id unique; registration_method in list; one event per user_id
Deduplication	unique on user_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Count new users and acquisition mix
KPI dependency	K-U01, K-U02, K-U10
Acceptance test	When a new account is created, exactly one user_registered event with a new user_id is recorded; repeat submission of the same form creates no second event.

REQ-002 user_logged_in

Item	Specification
Trigger	Authenticated session is created
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); session_id (string), device_type (enum), platform (enum)
Optional	login_method (enum)
Validation	session_id unique per login; user must exist
Deduplication	unique on (user_id, session_id)
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Returning-user and active-user KPIs
KPI dependency	K-U03, K-U04, K-U06
Acceptance test	Each successful authentication produces one event carrying a session_id; a refresh without re-authentication produces none.

REQ-003 task_started

Item	Specification
Trigger	Task attempt created
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); task_activity_id (string), task_id (string), task_type (enum), feature (enum), session_id (string)
Optional	provider_ref (string)
Validation	one task_activity_id per attempt
Deduplication	unique on task_activity_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Starts and completion rate denominator
KPI dependency	K-E01, K-E03
Acceptance test	Starting an attempt records one event with a unique task_activity_id before any completion event exists for it.

REQ-004 task_completed

Item	Specification
Trigger	Completion has passed the applicable validation process
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); task_activity_id (string), task_id (string), task_type (enum), feature (enum), status (enum: completed), validation_status (enum), reward_amount (decimal), reward_unit (enum)
Optional	provider_ref (string), duration_seconds (int)
Validation	exactly one event per task_activity_id; completion_ts >= start_ts; reward fields present when reward applies
Deduplication	unique on task_activity_id (idempotency key); replays ignored
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Primary earning outcome
KPI dependency	K-E03, K-E04, K-R01
Acceptance test	When a valid task completion occurs, exactly one analytically usable

Item	Specification
	completion event is recorded for the relevant task/user attempt, subject to the final engineering implementation and idempotency design.

REQ-005 task_failed

Item	Specification
Trigger	Attempt ends with an error or rejected validation
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); task_activity_id (string), task_id (string), failure_category (enum, Proposed list), failure_stage (enum: tracking, validation, availability, system)
Optional	failure_detail (string, no PII)
Validation	failure_category in approved list; one terminal event per attempt
Deduplication	unique on (task_activity_id, terminal state)
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / High
Why needed	Failure analysis
KPI dependency	K-E05, K-E06
Acceptance test	A failed attempt records one failure event with a failure_category from the approved list; no completion event exists for that task_activity_id.

REQ-006 reward_credited

Item	Specification
Trigger	Reward is written to the wallet ledger
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); reward_id (string), source_type (enum), source_ref (string), amount (decimal >0), unit (enum), wallet_transaction_id (string)
Optional	task_activity_id (string)
Validation	amount>0; wallet_transaction_id exists; one per reward_id
Deduplication	unique on reward_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Value delivered; reconciliation
KPI dependency	K-R01, K-R02, K-R05
Acceptance test	Every wallet credit from a reward has one reward_credited event whose amount equals the wallet ledger entry; no event is emitted for rewards still pending.

REQ-007 reward_reversed

Item	Specification
Trigger	Previously credited or pending reward is cancelled
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); reward_id (string), original_reward_id (string), amount (decimal), unit (enum), reason_category (enum), actor_type (enum)
Optional	case_ref (string)
Validation	original_reward_id must exist and be credited
Deduplication	unique on (reward_id, original_reward_id)
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / High
Why needed	Integrity and third-party reversal rate
KPI dependency	K-R04
Acceptance test	Reversing a reward produces one event referencing the original reward_id and a matching reversal ledger entry.

REQ-008 referral_converted

Item	Specification
Trigger	Referral meets conversion definition (A013)
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); referral_id (string), referrer_user_id (string), referred_user_id (string), conversion_rule (string)
Optional	reward_id (string)
Validation	after referral_activated; at most one per referral_id
Deduplication	unique on referral_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Conversion KPI
KPI dependency	K-F04
Acceptance test	A referral meeting the agreed conversion definition produces exactly one event; registration alone never produces it.

REQ-009 wallet_transaction_created

Item	Specification
Trigger	Ledger entry is created
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); transaction_id (string), transaction_type (enum), amount (decimal), unit (enum), status (enum), reference_id (string)
Optional	related_transaction_id (string)
Validation	transaction_id unique; amount != 0
Deduplication	unique on transaction_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Ledger completeness
KPI dependency	K-W01
Acceptance test	Each ledger write creates one event with a unique transaction_id; retries do not create new IDs.

REQ-010 wallet_transaction_completed

Item	Specification
Trigger	Ledger entry becomes final
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); transaction_id (string), transaction_type (enum), amount (decimal), unit (enum), balance_after (decimal)
Optional	-
Validation	transaction_id exists with status created
Deduplication	unique on transaction_id + status
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Credit volume, redemptions
KPI dependency	K-W02, K-W03
Acceptance test	A settled transaction records one completed event; the sum of completed credits less reversals equals the balance change for a test user.

REQ-011 wallet_transaction_reversed

Item	Specification
Trigger	Ledger entry is reversed
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); transaction_id (string), related_transaction_id (string), amount (decimal), unit (enum), reason_category (enum)
Optional	-
Validation	related_transaction_id must exist and be completed
Deduplication	unique on transaction_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Reversal rate
KPI dependency	K-W04
Acceptance test	A reversal references an existing completed transaction through related_transaction_id.

REQ-012 withdrawal_requested

Item	Specification
Trigger	Request is submitted and recorded
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); withdrawal_id (string), amount (decimal), unit (enum), method (enum, list pending A010)
Optional	-
Validation	amount>0; one per withdrawal_id
Deduplication	unique on withdrawal_id
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Request volume
KPI dependency	K-D01, K-D07
Acceptance test	Submitting a payout request records one event with a unique withdrawal_id and amount>0.

REQ-013 withdrawal_completed

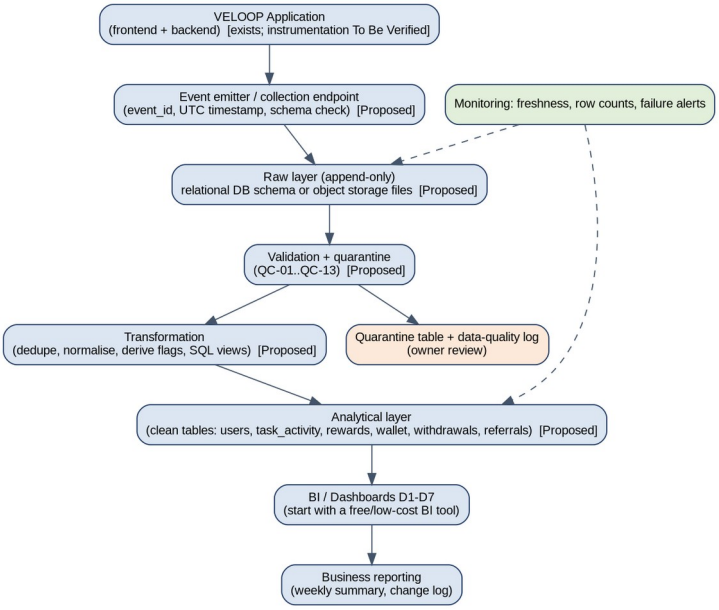
Item	Specification
Trigger	Payout confirmed
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); withdrawal_id (string), amount (decimal), unit (enum), method (enum)
Optional	payout_reference (string, tokenised)

Item	Specification
Validation	terminal; exclusive with rejected
Deduplication	unique on withdrawal_id + terminal status
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / Critical
Why needed	Completion rate, volume
KPI dependency	K-D02, K-D04, K-D06
Acceptance test	Completion appears once per withdrawal_id and never together with a rejected event.

REQ-014 withdrawal_rejected

Item	Specification
Trigger	Request refused
Payload (required)	user_id (string); event_id (UUID); event_ts (UTC ISO-8601); withdrawal_id (string), reason_category (enum)
Optional	-
Validation	terminal; exclusive with completed
Deduplication	unique on withdrawal_id + terminal status
Timestamp	UTC, server-generated when the state is committed
Source / Priority	Backend / High
Why needed	Rejection reasons
KPI dependency	K-D04
Acceptance test	Rejection records one event with a reason_category from the approved list and no completion event.

16 Data Pipeline and Operating Model (O)



Element	Practical proposal (Proposed; fits current stack once A030 is answered)
Source systems / producers	VELOOP application frontend and backend; third-party callbacks; support tool (if any)
Collection	Single event endpoint or server-side logger; schema check; event_id and UTC timestamp assigned once
Storage and raw layer	Start with the existing relational database or file storage; raw tables append-only and never edited
Validation	Rules QC-01..QC-13 on load; failing rows go to a quarantine table
Transformation	Scheduled SQL views/jobs: deduplicate, normalise, derive flags
Analytical layer	Clean tables matching the logical model; documented KPI views
Dashboard and refresh	Free or low-cost BI tool; daily refresh, hourly for the withdrawal queue
Monitoring and failure handling	Freshness and row-count checks; alert on failed load; retry; no silent drops
Ownership	Engineering: events and raw layer. Data analyst: validation rules, KPI views, dashboards. Product/Operations: definitions and review of quarantine

Deliberately not recommended at this stage: a dedicated data warehouse, streaming platform or large orchestration stack. Scale up only when event volume or query time justifies it.

17 Privacy, Access and Governance (P)

Role	Access (Proposed)
Engineering	Technical event implementation and raw-layer troubleshooting; no routine access to business reports needing personal data
Data Analyst	Approved analytical datasets with pseudonymous IDs; hashed email only if required
Product / Operations	Aggregated business reporting and dashboards; row-level views only for operational cases
Content Team	Aggregated content/help performance only
Finance	Wallet and withdrawal aggregates and reconciliation reports
Intern	Only approved, sanitised, clearly labelled datasets

Information that must not be unnecessarily exposed or stored in analytics: passwords, authentication secrets, API keys, payment credentials, private credentials, sensitive security controls and fraud thresholds. Other principles: data minimisation (full_name and email excluded from the analytics layer), role-based access, simulated and production data never mixed, consent respected (marketing opt-in only with consent Accepted, QC-17), retention period to be decided (A025).

18 Analytics Change Management (Q)

Change type	Process
Feature changes	Update coverage matrix and event plan; version bump; notify KPI owners
Event changes	Never rename silently: add new event, deprecate old for a defined overlap period, log in change log
Reward mechanic changes	Record effective date; add reward_rule_version; split trend charts at the change
Withdrawal method changes	Add to approved method list; keep history; annotate dashboards
Field renames	Keep mapping old to new in dictionary; transformation layer exposes both until retired
KPI definition changes	New version; restate history or mark break; list affected dashboards
Dashboard changes	Version the spec; record visuals added/removed; owner sign-off

The Metric/Event Change Log (workbook sheet 19) records date, change, old and new definition, reason, affected KPIs, dashboards and reports, owner and version. Baseline entries for v1.0 are included.

19 Analytics Readiness and Gap Assessment (R)

Dimension	Design (0-5)	Verified (0-5)	Gap priority	Reason
Event coverage	4	0	Critical	Without verified events no production KPI can be produced.
Data model	4	0	Critical	Wallet transaction_id and referral linkage unverified.
Data dictionary	4	0	High	Dictionary must be reconciled with the real schema.
KPI definitions	4	0	High	Many KPIs depend on tables not supplied (J3).
Data quality	3	0	High	Duplicates and malformed dates exist in sample.
Dashboard readiness	3	0	Medium	Cannot be built until data layer exists.
Engineering instrumentation	2	0	Critical	Instrumentation determines everything else.
Access control	3	0	Medium	Minimise personal data in analytics layer.
Documentation	4	0	Medium	Needs ownership and review cadence.
Reporting	2	0	Medium	Depends on pipeline.

Scoring note: "Design" rates how complete this project's specifications are (average 3.3). "Verified" rates what is confirmed in production (0, because no production evidence was supplied). Priorities are justified, for example: Critical - without a stable transaction ID, reliable wallet transaction deduplication cannot be guaranteed.

20 Assumption and Verification Register (S)

ID	Unknown	Why it matters	Risk	Owner	Status
A001	Does the backend currently emit task completion events?	Required for all task analytics and reward reconciliation	High	Engineering	To Be Verified
A002	Is there a stable, immutable user_id issued at registration (independent of email)?	Cross-event joins and de-duplication depend on it	Critical	Engineering	To Be Verified
A003	Will the program team supply the missing Task Activity, Rewards, Wallet, Withdrawals, Referrals, Feature Events sheets?	Needed to test relationships and KPIs required by J2/J3 on the intended dataset	High	Program team / Admin	To Be Verified
A004	Does production generate a session_id and where is a session closed?	Session-level KPIs and funnels need it	High	Engineering	To Be Verified
A005	Is event_id generated client-side or server-side, and is it idempotent on retry?	Duplicate events inflate KPIs	Critical	Engineering	To Be Verified
A006	Which of the 15 feature names actually exist in production today?	Event plan and dashboards must not cover non-existent features	Medium	Product	To Be Verified
A007	Is reward crediting synchronous with task completion or delayed by validation?	Determines whether reward_pending is a real state	High	Product / Engineering	To Be Verified
A008	What are the exact reward units (VE, XP, Gem, Token) and are there conversion rules?	Amount fields need a unit; cross-unit totals are invalid without rules	High	Product	To Be Verified
A009	Is there a single wallet ledger or one balance per currency?	Defines the transaction model and balance reconciliation	Critical	Engineering	To Be Verified
A010	Which withdrawal methods, minimums and review steps exist?	Method distribution, processing-time and rejection KPIs depend on it	High	Product / Operations	To Be Verified
A011	Is a status-change history stored for	Without history, activity before/after	High	Engineering	To Be Verified

ID	Unknown	Why it matters	Risk	Owner	Status
	accounts (e.g. Active to Suspended)?	suspension cannot be separated			
A012	Which external providers return completion/validation signals and with what delay?	Third-party delay and reversals affect reward timing and KPIs	High	Product / Engineering	To Be Verified
A013	How is a referral defined as "activated" and "converted"?	Referral funnel stages cannot be computed without a business definition	High	Product	To Be Verified
A014	Does the referral record capture the referring user (referrer user_id)?	Simulated Users only holds referral_source_detail = Friend/Family and a code per user	Critical	Engineering	To Be Verified
A015	What time zone are timestamps stored in?	Day-level KPIs shift if UTC vs IST is mixed	High	Engineering	To Be Verified
A016	What is the reference timestamp for days_since_registration and the active flags?	Derived fields could not be reproduced for a few users	Medium	Data owner of the dataset	To Be Verified
A017	Is is_active based on login, any event, or a qualifying activity event?	KPI definition of DAU/WAU/MAU	High	Product / Analytics	To Be Verified
A018	Is "platform" = Web a delivery channel while primary_device = OS family?	Both fields exist; platform has a single value in the sample	Low	Engineering	To Be Verified
A019	Can a Pending Verification account earn rewards?	Task02 sample shows tasks for such accounts	High	Product	To Be Verified
A020	Can a Suspended account keep earning/withdrawing?	Same, affects reversal and fraud-review reporting	High	Product / Operations	To Be Verified
A021	Is consent captured as a timestamped event or a current flag only?	Privacy audit needs history	Medium	Legal / Engineering	To Be Verified
A022	Which user fields are legally necessary (name, email, age, gender, city)?	Data minimisation for the analytics layer	High	Product / Legal	To Be Verified
A023	Will support tickets / help-center content be tracked in-product or in an external tool?	Support KPIs source system	Medium	Operations	To Be Verified
A024	Is an exchange (VE to Gem/Token) a wallet transaction type?	Transaction-type list in D1	Medium	Product	To Be Verified
A025	How long are raw events retained?	Storage cost and reprocessing capability	Low	Engineering	To Be Verified
A026	Does onboarding have defined milestones?	onboarding_milestone_reached cannot be specified without them	Medium	Product	To Be Verified
A027	Are reward reversals and corrections performed manually by staff or automatically?	Determines the actor field and audit needs	Medium	Operations	To Be Verified
A028	What is the intended meaning of lifetime_sessions = 0 with onboarding Completed (23 users)?	Possible data defect or onboarding completed outside a session	Medium	Dataset owner	To Be Verified
A029	Are mobile apps planned in addition to the web platform?	platform / app_version fields	Low	Product	To Be Verified
A030	Is there an existing analytics tool or warehouse?	Pipeline recommendation must fit current stack and cost	Medium	Engineering	To Be Verified

21 Implementation Backlog

Priority	Requirement	Reason	Dependency	Owner	Stage
Critical	Stable, immutable user_id issued at registration	Cross-event analysis (A002)	Authentication	Engineering	Prelaunch
Critical	Unique event_id with idempotent writes	Duplicate events inflate KPIs (10	Event service	Engineering	Prelaunch

Priority	Requirement	Reason	Dependency	Owner	Stage
		duplicates in sample audit)			
Critical	Wallet transaction_id and append-only ledger	Reconciliation and deduplication	Wallet	Engineering	Prelaunch
Critical	Referral table with referrer and referred IDs and stage timestamps	Referral conversion cannot be computed otherwise	Registration, Referral	Engineering	Prelaunch
Critical	Withdrawal lifecycle with requested, processing, final timestamps	Processing-time and queue KPIs	Withdrawals	Engineering	Prelaunch
Critical	Emit task_started / task_completed / task_failed events	Core earning analytics	Tasks	Engineering	Prelaunch
Critical	Emit reward_credited linked to ledger entry	Separate activity from credit	Rewards, Wallet	Engineering	Prelaunch
High	Account Status History table	Separate pre/post suspension activity	Accounts	Engineering	Prelaunch
High	Session identifier and session end rule	Funnels and session KPIs	Frontend/Backend	Engineering	Prelaunch
High	Approved enum lists (failure_category, reason_category, unit, method)	Prevents free-text drift	Product decisions	Product	Prelaunch
High	Timezone standard (UTC storage, IST reporting if needed)	Day-boundary KPIs	Event service	Engineering	Prelaunch
High	Confirm eligibility rules for pending and suspended accounts	Resolves DQ-017, DQ-026	Policy	Product / Operations	Prelaunch
High	Raw layer + validation job + quarantine table	Quality controls	Storage	Data / Engineering	Prelaunch
High	Publish KPI definitions and DAU qualifying-event list	Reproducible metrics	Product	Analytics	Prelaunch
High	Obtain missing Task 01 simulated sheets (A003)	Complete J2/J3 testing	Program team	Admin	Immediate
Medium	Consent history capture	Compliance audit	Legal	Engineering	Prelaunch
Medium	Third-party outcome fields (provider_ref, provider_status)	Attribute external effects	Providers	Engineering	Postlaunch
Medium	Data-quality dashboard tile and alerts	Operational trust	Pipeline	Data	Postlaunch
Medium	Role-based access and hashed identifiers in analytics layer	Privacy	Governance	Data / Engineering	Prelaunch
Medium	Support ticket category tracking	Support KPIs	Operations	Operations	Postlaunch
Low	Badge/level/ leaderboard analytics	Low business risk	Core events	Analytics	Postlaunch
Low	Advanced management dashboard	Needs production data	Production data	Analytics	Postlaunch

22 Deliverables Index and Definition of Done

No.	Deliverable	Where
01	Executive Summary	Report section 01
02	Product Data Audit	Report section 03 + workbook sheets 3-5
03	Data Coverage Matrix	Workbook sheet 4
04	User Analytics Specification	Workbook sheet 6, report section 04
05	Earning & Reward Analytics Specification	Report section 05
06	Wallet & Withdrawal Analytics Specification	Report section 06
07	Referral & Engagement Analytics	Report section 07
08	Event Tracking Plan	Workbook sheets 7-8
09	Data Dictionary	Workbook sheet 10
10	Logical Data Model	Report section 10 (ER diagram)
11	KPI Catalogue	Workbook sheet 11
12	Simulated Data Profiling	Workbook sheets 12-15, report section 12
13	Data Quality Audit	Workbook sheets 16-17, report section 13
14	Dashboard Blueprint	Report section 14, workbook sheet 18
15	Developer Analytics Handoff	Report section 15, workbook sheet 9
16	Data Pipeline Proposal	Report section 16
17	Privacy & Governance Proposal	Report section 17
18	Analytics Readiness Assessment	Report section 19, workbook sheet 20
19	Assumption & Verification Register	Workbook sheets 1-2
20	Implementation Backlog	Workbook sheet 21
21	Final Consolidated Report	This document

Definition of Done - self-check

Criterion	Status	Note
Product data areas mapped	Done	31 coverage rows
Analytical events defined	Done (Proposed)	42 events
User, reward, wallet/withdrawal, referral analytics documented	Done	Sections 04-07
Event naming standardised	Done	object_action
Data dictionary complete	Done	106 fields
Logical data model complete	Done	10 entities
KPIs reproducible	Done	54 KPIs
Simulated data profiled	Partly	Users fully; wallet/withdrawal/reward/referral sheets not supplied (A003)
Data quality problems identified	Done	26 issues
Controls proposed	Done	22 rules
Dashboard requirements documented	Done	44 visuals
Engineering requirements testable	Done	Acceptance test per event
Pipeline, governance documented	Done	Sections 16-17
Unknowns recorded	Done	30 assumptions
Gaps prioritised, backlog exists	Done	14 gaps, 22 items
Simulated data labelled; no fictional result as VELOOP result	Done	Header, wording rule
Documents internally consistent	To be reviewed	Counts in this report are generated from the workbook content; please re-check after any edit

This report is an analytical proposal. It makes no claim about the existence, performance or success of any VELOOP feature.